Sahaj Bhadja

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Summary

Software engineer with **2+ years of experience** building secure backend and workflow systems, writing and debugging production code, and delivering tested software in **Agile** environments with strong ownership of validation, documentation, and release quality.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built and debugged backend modules for a **multi-tenant AI education platform** on **Azure SQL, PostgreSQL, and Blob Storage**, supporting **500+ users** while reducing dashboard latency from **1.2s to 350ms**.
- Implemented validation checks, async job orchestration, and release workflows across **FastAPI** services and relational databases, processing **1K+ AI jobs** and **5K+ monthly integration jobs** with **99%+ reliability**.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed and tested a **Flask + React** grading platform that evaluated **10K+ submissions**, adding secure file-ingestion, rubric, and feedback workflows for instructor-facing operations.
- Collaborated on backend APIs, multithreaded processing, and **GitHub Actions CI/CD**, handling **10K+ grading requests/day** for **150+ instructors** and reducing turnaround time by **60%**.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Wrote and debugged modules for a cross-platform **microservice** data-sharing system, automating build, install, and integration test flows across **Windows, Linux, and macOS** VMs with **Jenkins, PyTest, and Mockito**.
- Supported documentation, ETL-style API workflows, and Agile sprint delivery with Microsoft/GitHub stakeholders, improving response times by **30%** and pipeline efficiency by **68%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built features for a **MERN**-based resource management platform with **REST APIs**, **WebSockets**, and Azure SQL-backed services, synchronizing **2M+ records** across distributed business workflows.
- Supported automated testing, Docker packaging, and Jenkins deployments while improving platform efficiency, reducing resource wastage by **42%**, and increasing lead conversions by **15%**.

Projects

Chat2SQL Repo | *Java, TypeScript, SQL, Docker*

- Extended a **Java** and TypeScript database workspace to support **10+ databases**, schema exploration, and AI-assisted query authoring, with cross-platform setup through **Maven**, Node.js, Yarn, and Docker.

HMIS - Hospital Management Information System Live | *MERN, TypeScript, Twilio*

- Built a secure healthcare workflow platform with consent-based record access and appointment automation, supporting **100+ daily appointments** and reducing administrative workload by **30%**.

CPU-Scheduling Simulator Live | *C++, React.js, EJ-2*

- Implemented core scheduling algorithms and tested edge cases across **1K+ simulated process combinations**, reducing debugging time by **20%** and average turnaround time by **25%**.

Skills

Languages: Java, Python, C++, SQL, JavaScript/TypeScript
Development: Flask, FastAPI, Node.js, React, REST APIs, OOP, Debugging, System Design
Data & Tools: PostgreSQL, MySQL, Azure SQL, AWS, Maven, Docker, Git/GitHub, Jenkins
Engineering Practices: Agile, CI/CD, Unit Testing, Integration Testing, Documentation, Performance Optimization